

Homework 10 - Results

Unformatted Output

Running Task 1: Single GPU Image convolution

H2D Data-Transfer: 1.05905533ms
GPU-Avg-Time (convolution) global memory: 2.03315711ms
D2H Data-Transfer: 0.80490112ms

Running Task 2: 100 sequential GPU Image convolution

H2D avg Data-Transfer: 1.05138302ms
H2D all Data-Transfer: 105.13830185ms

GPU-Avg-Time convolution sequential: 4.37034607ms
GPU-All-Time convolution sequential: 437.03460693ms

D2H avg Data-Transfer: 0.69632292ms
D2H all Data-Transfer: 69.63229179ms

Time for all (sequential): 611.80520058ms

Running Task 3: 100 concurrent GPU Image convolution

All-Time (Avg per Image): 4.10706997ms
All-Time (All Images): 410.70699692ms
Success!

Task 1: Execution Time Measurements (Full HD: 1920 x 1080)

Operation	Execution Time
Host-to-Device transfer	1.06 ms
Kernel execution	2.03 ms
Device-to-Host transfer	0.80 ms

Performance Summary

Implementation	Average Time	Speedup
GPU - no streaming	611.81 ms	baseline
GPU - with streaming	410.71 ms	1.49x